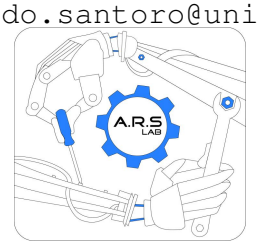
# System Identification

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## Notes

# System Identification

## System Identification

**System Identification** is a process that aims at deriving the model of a system from the measurement of input and output data.

## Approaches

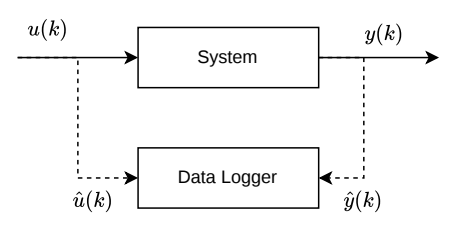
- **Black-box model**, you know nothing about the internal structure of the system
- **White-box model**, you know everything, thus you can derive the model from e.g. physical laws
- **Grey-box model**, you know something, e.g. the expected behaviour, thus you need to derive the parameters



Notes

# System Identification

## Step 1. Collect measures




- Send to the system an input signal (preferably a canonical one)
- By using a data logger, sample input and output signals, $\hat{u}(k)$ and $\hat{y}(k)$, with a sampling time $\Delta T$ able to capture the dynamics of the system

We have:

$$\begin{aligned} \hat{u}(k) &= u(k) + \beta \\ \hat{y}(k) &= y(k) + \gamma \end{aligned}$$

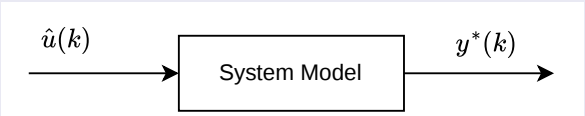
were $\beta$ and $\gamma$ are the measure noises, i.e. random gaussian variables with mean = 0



Notes

## Step 2. Model Derivation

- By means of a proper methodology/algorithm derive a suitable model of the system




that minimizes the mean square error:

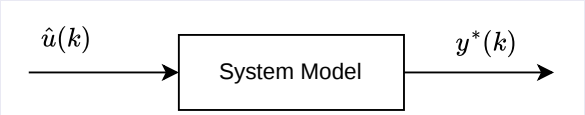
$$\min E(y^* - \hat{y})^2$$



Notes

## Step 3. Model Test

- Test the model and compare its output $y^*$ with the measures $\hat{y}$






Notes

# Least Squares Method



Notes

## The ARX Model

Given an ARX model:

$$y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

where $n$ and $m$ are known, the aim is to find:

- $a_1, \ldots, a_n$ parameters
- $b_0, \ldots, b_m$ parameters

The method is based on using measure samples to build the output vector and the measure matrix, and use them to compute paramaters.



Notes

## Least Squares Method

Let's choose $n = 2$ and $m = 1$. We have:

$$\begin{aligned} y(2) &= -a_1 y(1) - a_2 y(0) + b_0 u(2) + b_1 u(1) \\ y(3) &= -a_1 y(2) - a_2 y(1) + b_0 u(3) + b_1 u(2) \\ y(4) &= -a_1 y(3) - a_2 y(2) + b_0 u(4) + b_1 u(3) \\ &\dots \end{aligned}$$

In matrix form:

$$\begin{bmatrix} y(2) \\ y(3) \\ y(4) \\ \dots \end{bmatrix} = \begin{bmatrix} y(1) & y(0) & u(2) & u(1) \\ y(2) & y(1) & u(3) & u(2) \\ y(3) & y(2) & u(4) & u(3) \\ \dots & & & \end{bmatrix} \begin{bmatrix} -a_1 \\ -a_2 \\ b_0 \\ b_1 \end{bmatrix}$$

In other words:

$$Y = \Phi P$$

The aim is to find $P$



Notes

## Least Squares Method

In matrix form:

$$\begin{bmatrix} y(2) \\ y(3) \\ y(4) \\ \cdots \end{bmatrix} = \begin{bmatrix} y(1) & y(0) & u(2) & u(1) \\ y(2) & y(1) & u(3) & u(2) \\ y(3) & y(2) & u(4) & u(3) \\ \cdots & & & \end{bmatrix} \begin{bmatrix} -a_1 \\ -a_2 \\ b_0 \\ b_1 \end{bmatrix}$$

In other words:

$$Y = \Phi P$$

- $Y$ is the output vector
- $\Phi$ is the measure matrix
- $P$ is the parameter vector

The aim is to find $P$



Notes

## Least Squares Method

In matrix form:

$$Y = \Phi P$$

The equation cannot be inverted since $\Phi$ is not square.

We can apply the least squares method as follows:

$$\Phi^T Y = \Phi^T \Phi P$$

Now $\Phi^T \Phi$ is a square matrix and can be inverted, so:

$$(\Phi^T \Phi)^{-1} \Phi^T Y = P$$

Hence the solution



Notes

# Neural ARX Method



Notes

## The ARX Model

Given an ARX model:

$$y(k) + a_1 y(k-1) + a_2 y(k-2) + \cdots + a_n y(k-n) = b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

rewritten as:

$$y(k) = -a_1 y(k-1) - a_2 y(k-2) - \cdots - a_n y(k-n) + b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

where $n$ and $m$ are known, the aim is to find an function block that gives $y(k)$ from inputs:

- $y(k-1), \ldots, y(k-n)$
- $u(k), \ldots, u(k-m)$



Notes

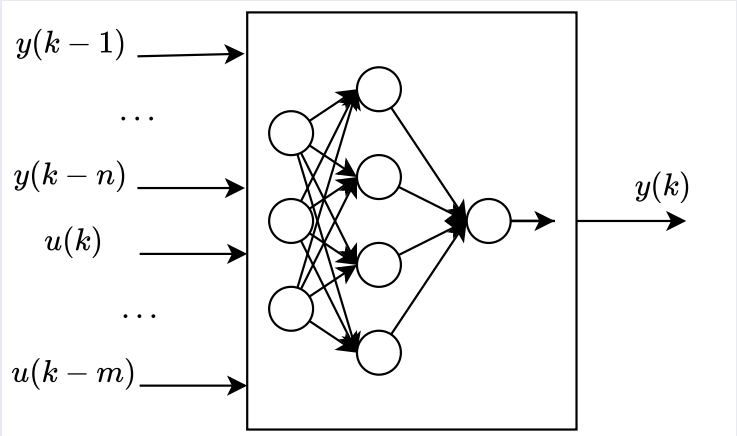
# Neural ARX Method

## The ARX Model

Given an ARX model:

$$y(k) = -a_1 y(k-1) - a_2 y(k-2) - \cdots - a_n y(k-n) + b_0 u(k) + b_1 u(k-1) + \cdots + b_m u(k-m)$$

The method is based on building a MLP (Multi-Layer Perceptron) and train it using the data sampled:



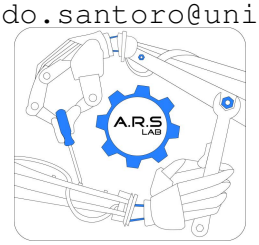



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